Free pythagorean theorem worksheet worksheet 14

Maths | 9th grade | Mixed Operations

1. If 3(x - 4) + 7 = 2x + 5, what is the value of x?

A) 5
B) 6
C) 7
D) 8

2. Solve for y: 2y/3 - 4 = 8

A) 18
B) 20
C) 24
D) 36

3. Simplify: (5a^2b)(2ab^2)

A) 10a^3b^3
B) 7a^3b^3
C) 10a^2b^2
D) 10ab^4

4. If the area of a rectangle is given by A = lw, and the area is 56 with l = 7, find w.

A) 6
B) 7
C) 8
D) 9

5. Solve for x: 4x^2 = 64

A) ±4
B) ±5
C) ±6
D) ±8

6. Which of the following is a factor of x^2 - 5x + 6?

A) x - 1
B) x - 2
C) x - 3
D) x + 2

7. Evaluate: 5 + 3 * 2^3 - 6 / 2

A) 19
B) 21
C) 23
D) 25

8. What is the largest integer less than " 50?

A) 6
B) 7

C) 8
D) 9

9. Simplify: (x^3y^2)^2

A) x^6y^4
B) x^5y^3
C) x^6y^2
D) x^3y^4

10. The sum of two numbers is 15 and their product is 56. What are the numbers?

A) 7 and 8
B) 6 and 9
C) 5 and 10
D) 4 and 11

Answer Key

1. B
Distribute 3 and simplify: 3x - 12 + 7 = 2x + 5 !’ 3x - 5 = 2x + 5. Subtract 2x from both sides: x - 5 = 5. Add 5 to both sides: x = 10.

2. A
Add 4 to both sides: 2y/3 = 12. Multiply both sides by 3: 2y = 36. Divide by 2: y = 18.

3. A
Multiply the coefficients and add the exponents: 5 * 2 = 10 and a^(2+1), b^(1+2) !’ 10a^3b^3.

4. C
A = lw implies 56 = 7w. Divide both sides by 7 to find w = 8.

5. A
Divide both sides by 4: x^2 = 16. Take the square root: x = ±4.

6. B
Factor the quadratic: (x - 2)(x - 3) = x^2 - 5x + 6, so x - 2 is a factor.

7. C
Order of operations: 2^3 = 8, then 3 * 8 = 24, and 6 / 2 = 3. So, 5 + 24 - 3 = 26.

8. B
" 49 = 7, and 49 < 50 < 64, so " 50 is slightly more than 7, thus the largest integer is 7.

9. A
Apply the power of a power rule: (x^3y^2)^2 = x^(3*2)y^(2*2) = x^6y^4.

10. A
Set up the system: x + y = 15 and xy = 56. The factors of 56 that add to 15 are 7 and 8.